

PigmentationAtlas transforms proximity-based GWAS annotation into evidence-based target-gene prioritization

Colocalization altered the conventionally assigned candidate gene in 111 of 130 evaluable pigmentation GWAS clumps (85.38%).

A

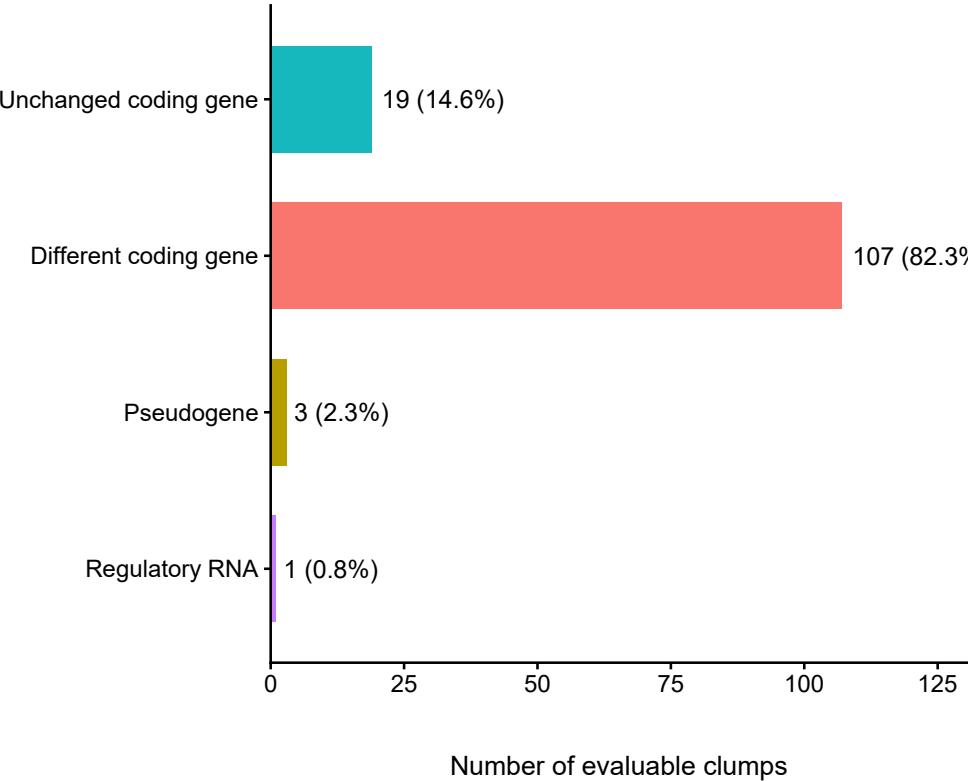
From genomic proximity to colocalization-supported target genes



Proximity-based annotation is replaced by evidence-based target-gene prioritization

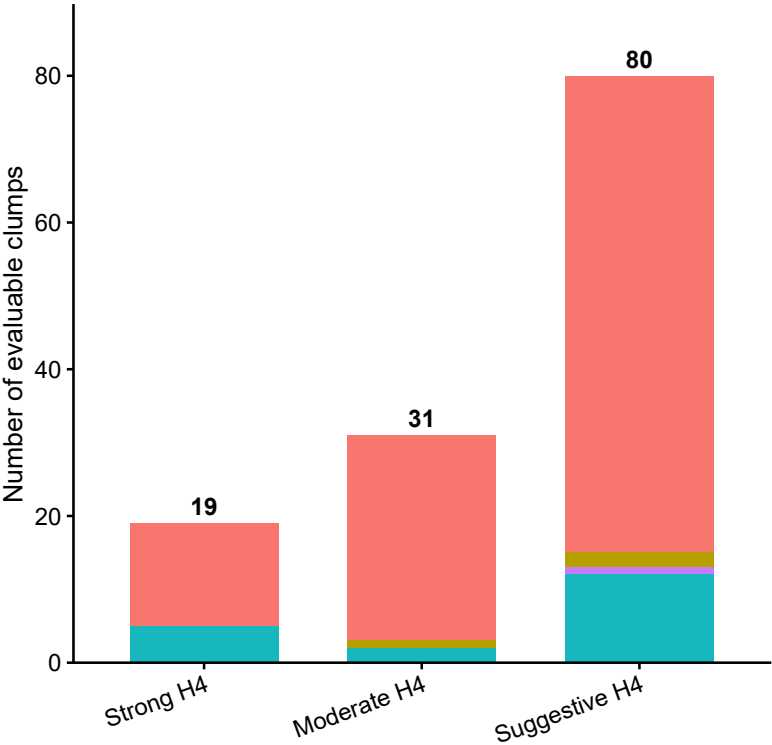
B

Biological outcome of candidate-gene reassignment



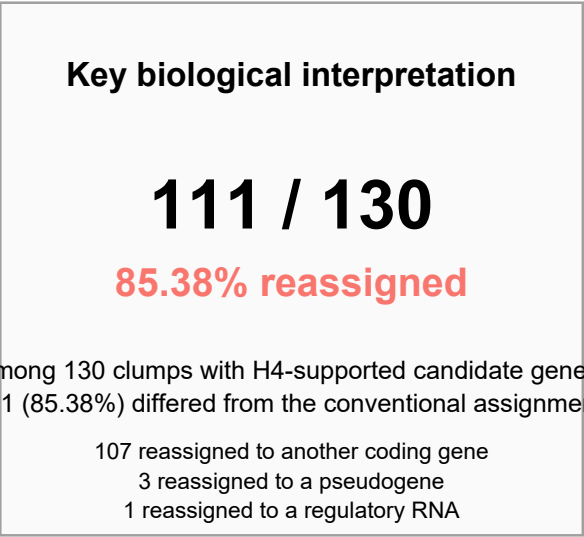
C

Reclassification is observed across all H4 evidence classes



D

PigmentationAtlas reveals widespread reinterpretation



Outcome Unchanged coding gene Different coding gene Pseudogene Regulatory RNA